

SmarPod Software Installation



www.smaract.com





Copyright © 2011 - 2024 SmarAct GmbH

Specifications are subject to change without notice. All rights reserved. Reproduction of images, tables or diagrams prohibited.

The information given in this document was carefully checked by our team and is constantly updated. Nevertheless, it is not possible to fully exclude the presence of errors. In order to always get the latest information, please contact our technical sales team.

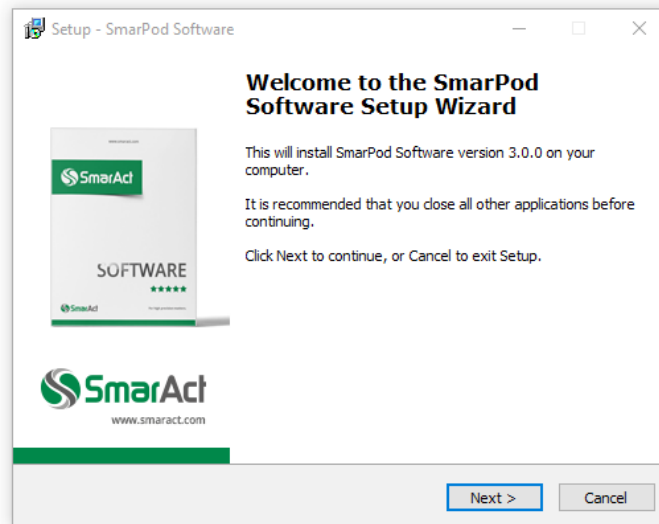
SmarAct GmbH, Schuette-Lanz-Strasse 9, D-26135 Oldenburg
Phone: +49 (0) 441 - 800879-0, Telefax: +49 (0) 441 - 800879-21
Internet: www.smaract.com, E-Mail: info@smaract.com

Document Version 1.11.0.0, Build 151

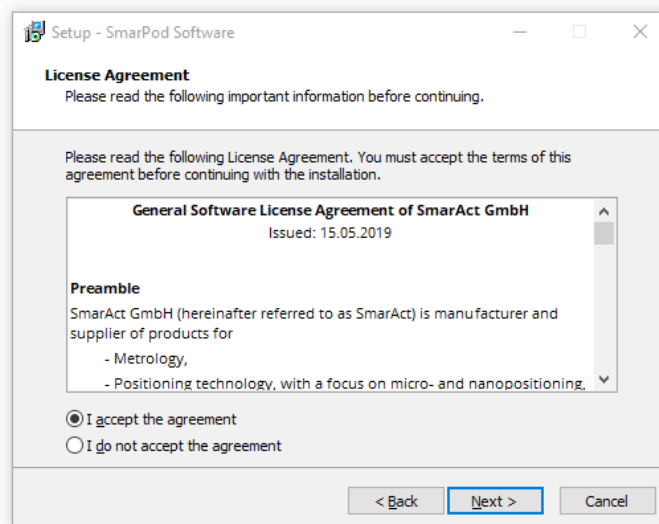
1 INSTALLING SMARPOD WINDOWS SOFTWARE

On the **SmarPod Software installation medium** you will find all software needed to use the *SmarPod* and to develop your own software. The installation medium is either a CD or a USB memory stick.

To start the installation run **SmarPod_Installer_x.y.z.exe**, where x.y.z is the version of the installer.



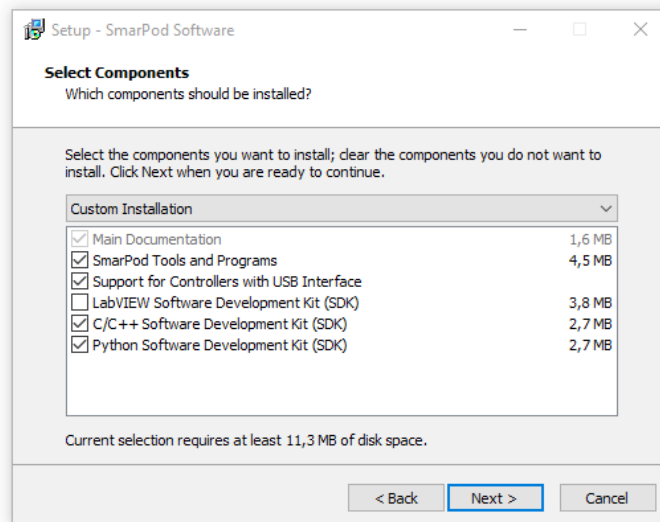
Read and accept the Software Licensing Agreement (EULA)



Select the location for the *SmarAct* software folder where most of the files will be installed (default is C:\SmarAct). If the software has been installed before, the installation location cannot be changed. Uninstall the old installation first in order to select a new location.

1.1 Selecting Components for Installation

On the *Select Components* page, select the components you wish to install:



If you select the *SmarPod Tools and Programs*, the *SmarPod Controller* GUI application is installed, which allows to move the SmarPod manually and with a Hand Controller (see *SmarAct Hand Controller*). The installer will show a warning that the *LabVIEW™ Runtime Engine* (32 bit) must be installed in order to use the software. You can find the installer on the installation medium in the *Drivers* folder. If a *LabVIEW™ 2018* (32 bit) development system is already installed on your PC, the *Runtime Engine* doesn't have to be installed.

Support for Controllers with USB Interface is selected by default. If selected, support for communicating with MCS controllers over USB cable will be installed. If it is unchecked, some 3rd party libraries for USB communication are not installed and the SmarPod software won't be able to communicate with controllers over USB.

1.1.1 Software Development Kits (SDK)

For writing your own software for the SmarPod, software development kits (SDK) are available for different programming languages:

- For C/C++ select the *C/C++ Software Development Kit (SDK)* option.
- For Python select the *Python Software Development Kit (SDK)* option.

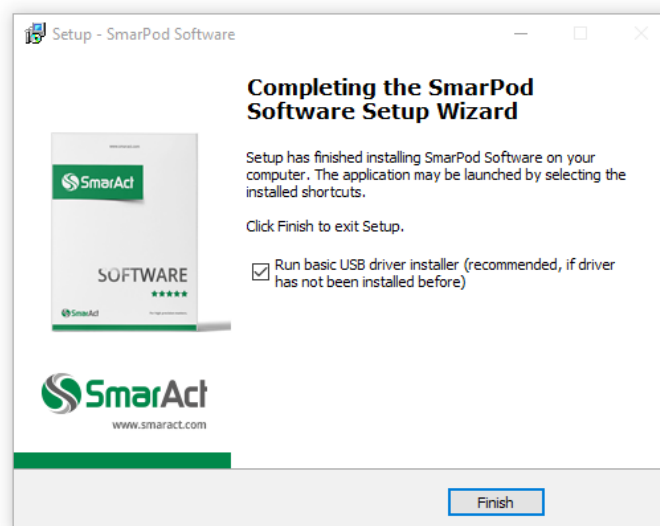
If this SDK is selected, the installer will ask in one of the following steps to install the API directly in your local Python installation. This requires that *Python 3.4* or later and the Python package installer *pip* are installed and that you have network access to the *PyPi* Python package repository to download some additional packages. (Direct installation of the package might not work if you have Python installed for your own user account only and not for all users). Alternatively you can choose to install just the package ZIP file which you can install in Python manually later. For more information see *Where Files are Installed* below and the document *Using SmarAct Python SDKs.pdf*.

- For LabVIEW™ select the *LabVIEW Software Development Kit (SDK)* option. Installation of this SDK requires a LabVIEW™ development system to be installed on the PC. If more than one version of LabVIEW™ is installed, choose the version in one of the following steps. To install for additional versions, run the installer again and select another version (the other components don't have to be installed again).

1.1.2 Installing the USB Interface Driver

Installation of the USB interface driver is required if you have a SmarAct controller with a USB interface.

If the SmarPod software installer was started from the installation medium and *Support for Controllers with USB Interface* was selected in the components selection step, the installer will ask if the 3rd-party *USB-to-serial* driver installer should be executed after the main files have been installed:



You can skip the driver installation by unchecking the checkbox. If not installed, it will not be possible to control SmarPods connected via USB. (The software will report a "Driver Error" in this case).

Alternatively, if the SmarPod software installer was not started directly from the installation medium or the option was not selected during installation, the USB interface driver may be installed by running the program *CDMxxxx_Setup.exe* from the *Drivers* folder on the installation medium (where *xxxx* is a number).

Also, if you install your own software on a different Windows PC and the SmarPod is connected via USB, the USB interface driver installer needs to be run there as well.

1.2 Where Files are Installed

The **SmarPod folder** is created inside the **SmarAct folder** which you have selected during installation. The default location of the SmarAct folder is *C:\SmarAct*.

Documentation is installed in the *Documentation* folder inside the SmarPod folder.

The **Programs** folder in the SmarPod folder contains **application software** like the *SmarPod Controller* GUI program.

Shortcuts to programs and documents are created in the Windows *Start* menu.

The **SDK** folder contains the software development kits that have been selected in the installer.

- **C/C++:** folder *SDK\C* contains sub folders for code examples, C header and library files. The path to this folder is in the environment variable *SMARPOD_SDK* which can be used in development tools to find the SDK without specifying the folder path directly. For example, in *Visual Studio*, you may add *\$(SMARPOD_SDK)\include* to the include paths of your project which enables the compiler to locate the *SmarPod.h* header if it is included by your C or C++ source files.
- **Python:** the folder *SDK\Python\examples* contains Python programming examples. If you have chosen the option to install the python package directly in your local Python installation (see *Installing SmarPod Windows Software*), the package **smaract.smarpod** will be available immediately in your Python interpreter. Under *SDK\Python\packages* you can find the Python API package as a ZIP file for manual installation with the *pip* tool. Refer to the *Using SmarAct Python SDKs.pdf* document for instructions.
- **LabVIEW™:** the SmarPod commands are installed in the LabVIEW™ program folder and can be selected directly from the VI palettes. Folder *SDK\LabVIEW\examples* contains the programming example VIs.
- **Redistributable files:** if any SDK has been installed, the folder *SDK\Redistributable* is created, which contains shared libraries (DLLs) needed to run SmarPod software on a computer where the installer was not run.

If the SmarPod is connected via USB interface, the USB interface driver must be installed on the target machine too. See *Installing the USB Interface Driver*.

1.3 Uninstalling the SmarPod Software for Windows

In the software installation folder you can find an uninstaller program in *SmarAct\SmarPod\Uninstall* which removes all SmarPod software from your disk. Other SmarAct software is not removed with the SmarPod uninstaller. Use the uninstaller for the other products to remove their software.

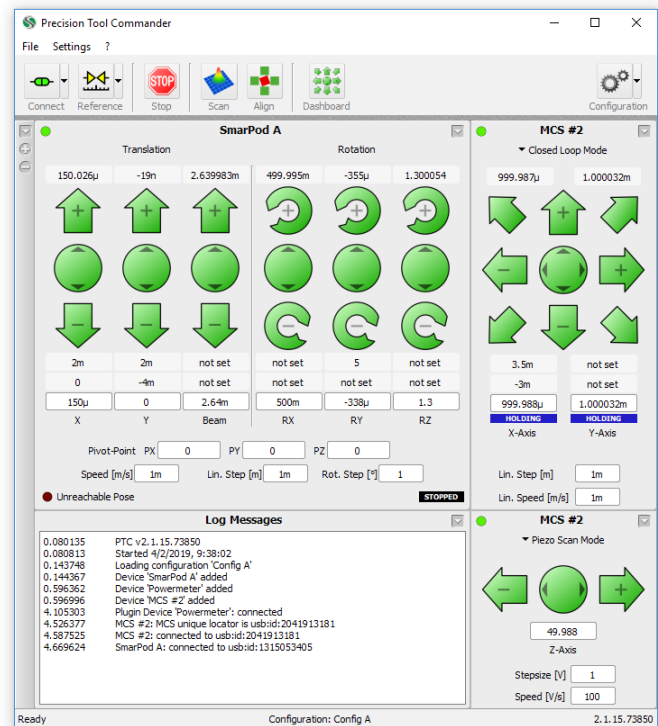
2 LINUX SOFTWARE

The C and Python SDKs are also available for the Linux operating system running under the x86 or x86_64 processor architectures. The SmarPod Linux software is available as a compressed *tar* archive on the installation medium: **SmarPod_x.y.z_Linux.tgz**, where x.y.z is the software version.

Please refer to the instructions in the **README** text file in that compressed archive for information about installing the software.

3 PRECISION TOOL COMMANDER (PTC)

Optionally the *Precision Tool Commander (PTC)* program can be installed under Windows. The application provides a graphical user interface for controlling positioners and performing basic measurements as well as automatic alignment tasks. *PTC* must be installed with the installer **PTC_Installer_x.y.z.exe** which can be found on the installation medium.



4 SMARACT HAND CONTROLLER

If you have a *SmarAct Hand Controller*, you can use it with the *SmarPod Controller* GUI without installing additional software except for the USB-to-serial driver (see above).

If you want to use the hand controller with your own software, you can find the installer **SmarActHID_Installer_x.y.z.exe** for the hand controller software development kit (SDK) in the *SmarActHID* folder on the installation medium.

